Technical Report: Periodic-1D Rank-Two Band Equivalence versus Hopping Locality

Project Koios

Development migration draft

1 Scope and evidence boundary

This report documents a controlled synthetic one-dimensional periodic calculation used to study the distinction between pointwise band equivalence and hopping locality. The full parent has four states per cell; the isolated retained space has rank two. The calculation is not a Mathieu equation, a material Hamiltonian, a density-functional calculation, or a scientific validation of a semiconductor model.

At this migration stage reusable numerical implementation remains with its Python owner, while calculation-specific development producers, tables, and figures have been extracted from the experiment into this calculation root. This report does not promote those development outputs into retained evidence. Historical scalar and M1–M3 packages retain their original identities and evidence boundaries.

2 Model and conventions

The parent is a one-dimensional finite-range Bloch Hamiltonian

$$K(q) = \sum_{R=-1}^{1} e^{2\pi i q R} T_R, \qquad -\tfrac{1}{2} \le q < \tfrac{1}{2}, \tag{1}$$

with four orbitals per cell and Hermitian reverse hopping $T_{-1} = T_{+1}^{\dagger}$. The two lowest eigenvectors are retained only when the second-to-third-band isolation gate passes. A discrete parallel-transport and closure procedure supplies a periodic orthonormal frame $F(q)$, giving

$$H(q) = F(q)^{\dagger} K(q) F(q). \tag{2}$$

The band-equivalent comparison field is

$$\theta_a(q) = a \sin(2\pi q), \tag{3}$$

$$U_a(q) = R(\theta_a(q)), \tag{4}$$

$$A_a(q) = U_a(q)^{\dagger} H(q) U_a(q), \qquad 0 \le a \le \pi/4. \tag{5}$$

Thus $A_a(q)$ and $H(q)$ have identical ordered eigenvalues at every momentum, while their Fourier hopping blocks in the declared retained coordinates generally differ. The comparison admits one momentum-independent real rotation and deliberately excludes reuse of $U_a(q)$.

3 Simulation method

The development calculation has three related layers. A flat traceless rank-two limit supports exact analytic correlation and residual identities. A dispersive four-state finite-range parent supplies a momentum-dependent retained subspace and transported periodic frame. A bounded parent homotopy separates representation-induced effects from pre-existing parent dispersion and hopping locality.

Momentum calculations assemble the same Bloch parent on nested meshes; refinement does not change the parent Hamiltonian. Fourier blocks are evaluated with an explicit shifted-mesh convention. Even finite-cyclic data are converted to an ordinary-lattice inventory by removing and separately reporting the centered Nyquist block. Ordered symmetric cutoffs are then evaluated on a disjoint momentum mesh. Cyclic conversion, finite interpolation, deliberate truncation, and total sorted-band effects remain separate channels.

The versioned five-stage calculation entry points have not yet been created. Existing values remain development outputs of the maintained experiment rather than results produced from this successor directory.

4 Independent verification

The maintained verifier reconstructs harmonic maximization, frame alignment, raw block powers, and selected observables without trusting producer diagnostics. Verification must check parent Hermiticity, retained-space isolation, frame orthonormality, reciprocal closure, projector reconstruction, pointwise spectral equivalence, Parseval identities, and range-selection decisions.

For the flat family, finite-mesh correlation and the analytic $J_0(2a)$ comparator are distinct fields. Agreement between them is a convergence diagnostic, not permission to replace finite data with the analytic value. For the dispersive family, shared parent eigenvalues and projectors supply frame-invariant refinement channels. Neighbor-overlap singular values are qualification gates rather than convergence observables.

No independent verification result is presently owned by this successor directory.

5 Development analysis and results

The existing development analysis reports three bounded observations. First, the smooth momentum-dependent rotation preserves the complete retained spectrum pointwise while redistributing Fourier hopping power. Second, the exact flat-parent residual–tail identity deforms systematically when non-onsite parent hopping is introduced. Third, on the declared sampled truncation inventory, reference and rotated representations can reach the same sampled band-error target at different first-passing cutoffs.

These observations remain development calculations. The momentum study is reported as converged on its declared nested meshes, whereas the capped uniform amplitude interpolation study remains unresolved. The range calculations are finite sampled decisions with a bounded cutoff inventory; they are not continuum extrema or universal range laws.

Numerical tables and figures will be incorporated only after the simulation and analysis wire contracts are migrated and independently verified from this calculation root.

6 Limitations and nonclaims

The model is synthetic and uses an arbitrary positive energy scale. It supplies no material parametrization, uncertainty quantification, thermodynamic-limit result, or continuum certificate. Finite momentum and amplitude samples do not establish extrema between samples. A first-passing cutoff is conditional on the declared mesh, evaluation set, norm, tolerance, and maximum tested range.

The restricted global $SO(2)$ alignment is not maximally localized Wannier gauge optimization. Pointwise unitary equivalence does not make two represented matrix fields identical under the declared restricted comparison. Conversely, representation-induced hopping redistribution is not evidence of a changed physical band spectrum.

Lean formalization of related finite identities does not certify NumPy execution, serialized JSON, plots, binary64 rounding, or the applicability of this model to a material system.

7 Provenance and reproduction

Calculation-specific workflow implementation and tests are owned under

```
calculations/periodic1d/rank2/band-equivalence-versus-hopping-locality/
  workflow/
  tests/
```

Generic reusable numerical capabilities remain with their canonical library owners. The publication-independent experiment specification is under

```
experiments/model-adequacy/periodic1d/
  band-equivalence-versus-hopping-locality/
```

Calculation-specific development entrypoints are at the calculation root; their inputs and tables are under

```
calculations/periodic1d/rank2/band-equivalence-versus-hopping-locality/
  simulations/development/
```

and plot sources and figures are under the sibling `analysis/development/` subtree. The working publication source is under

```
docs/publications/papers/Ragasa2026a/v3/
```

Historical retained packages remain at their original calculation paths.

A complete successor reproduction command cannot be stated until the versioned input, closed serializers, five-stage entry points, and retained-output policy are implemented. Development re-execution is replaceable workflow validation, not retained production evidence or scientific validation.